

LIFE ON MIR



IN ORBIT FROM 1986 TO 2001, the Mir space station was the site for the most extensive experiments in living outside the Earth's atmosphere that have yet been attempted. Russian cosmonaut and physicist Valeriy Polyakov established the world record for space endurance by spending 437.75 days on board Mir in 1994–95, while his colleague Sergei Avdeyev spent a record cumulative total of 747.6 days orbiting in space.

Of the American astronauts who joined the cosmonauts on Mir from 1995 to 1998, the longest stay was made by Shannon Lucid, who spent 188 days in orbit.

It was found that the human body adapted very quickly to gravity-free conditions. For example, the heart, veins, and arteries adjusted to the relative ease with which blood would circulate. Adapting back to conditions on Earth posed more serious problems. Some cosmonauts found that they were too weak to walk or even sit up unaided when returned to gravity after months of weightlessness.

To counter the disorientating effect of an environment in which the Sun rose and set 16 times in

FLOATING WORLD

Every task in space presented cosmonauts and astronauts with special problems related to weightless conditions. To sit on a chair, for example, they would have to hook their feet into rings on the floor.



MICHAEL FOALE

By the time that British-born American Mike Foale spent time on Mir in 1997, the aging station was becoming seriously unsafe.



SHANNON LUCID

American biochemist Shannon Lucid broke a string of records with her 188 days on board Mir in 1996, a trip prolonged by a problem with the Shuttle that delayed her return.

24 hours, a strict Earth-based time routine was imposed, with cosmonauts woken by a wrist alarm at 8:00am Moscow time every day. The crew slept in floating sleeping bags attached to a wall. There was no real sense of “up” or “down” – astronaut Jerry Linenger described sleeping with his feet to the ceiling and his head near the floor.

Life on board was inevitably cramped and nostalgia for friends and family affected all the crew. Mir was increasingly cluttered with objects brought up from Earth and never returned, including personal effects such as books and videos. Still, Lucid recalled the small pleasures of communal life – for example, the crew eating meals together “floating around a table in the Base Block,” with Jello as a Sunday treat. Later Americans on Mir had a darker experience as the station deteriorated, with systems breakdowns, power failures, and other life-threatening crises.

to economic and technological difficulties, and a lack of scientific and military support.

The failure of Soviet President Mikhail Gorbachev's attempts to modernize the communist state led to the political disintegration of the Soviet Union. At the end of 1991, the state that had once taken on the United States in the space race ceased to exist.

Running out of cash

The personnel and facilities of the Soviet space program were primarily inherited by the Russian Federation, although the Baikonur Cosmodrome was in independent Kazakhstan. Even if the successor states of the Soviet Union had the will to continue the space program – and to a degree they undoubtedly did – money was in desperately

short supply. With budgets slashed to a small fraction of those in the Soviet era, plans for a Mir 2 space station were scrapped and unfinished Burans were dismantled.

Even before the fall of the Soviet Union, the new spirit of enterprise sweeping the communist bloc had led the Glavkosmos space agency, set up in 1985, to sell journeys to Mir for millions of dollars at a time. For example, one place was bought by the Japanese for \$7 million in order to put one of its nationals into space. Instead, the American space agency incorporated Mir into its own program for building a new space station.

President Reagan's project for Space Station Freedom had, like so many bold manned space projects, run aground on the issue of cost. A way forward was found in international cooperation,

with Canada, Japan, and the countries of the European Space Agency soon signing up to take part. Adding Russia to the consortium promised a financial lifeline for the beleaguered ex-Soviet space researchers, engineers, and cosmonauts, while the use of Russian facilities had much to offer in terms of economy to their American counterparts. Cooperation across the previous Cold War divide also had a political payoff, symbolizing aspiration to a new world order.

As part of the International Space Station project, in 1994 the Russians and Americans began a series of joint missions in which cosmonauts traveled on Shuttles, Shuttles docked with Mir, and American astronauts spent time on board the Russian space station. Over five years, the old rivals learned to work together.